

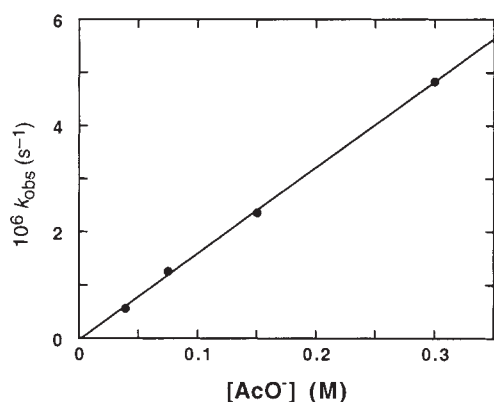


FIG. 4 Variation of the pseudo-first-order rate constant for 5-nitrobenz-isoxazole decomposition as a function of acetate concentration. The k_{obs} values were obtained by the method of initial rates in water (μ 0.10, pH 5.4, 20 °C) at a buffer ratio of 4:1 sodium acetate:acetic acid. The second-order rate constant measured at 30 °C under otherwise identical conditions ($k_{\text{AcO}^-} = 5.26 \times 10^{-5} \text{ M}^{-1} \text{ s}^{-1}$) fits the previously reported¹¹ Brønsted equation correlating the rates of deprotonation of **1** (Brønsted $\beta = 0.73$, correlation coefficient = 0.999).

thermore, simultaneous induction of acid-base pairs capable of additive bifunctional catalysis^{20,21} may provide even larger effects than are possible with a single base, allowing many transformations of interest to chemists and biologists, including pro-

cesses that are normally disfavoured²², to be accelerated with enzyme-like efficiency. □

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Ecosystem changes in the North Pacific subtropical gyre attributed to the 1991–92 El Niño

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SUBTROPICAL ocean gyres are considered to be the marine analogues of terrestrial deserts because of chronic nutrient depletion and low standing stocks of organisms¹. Despite their presumed low rates of primary and export production, oligotrophic habitats contribute significantly to global productivity because of their large extent². Therefore, even small changes in ecosystem production can produce large effects in the global carbon cycle. The North Pacific subtropical gyre has generally been thought to support a homogeneous, stable biological community^{3,4}, but recent investigations have suggested instead that the ecosystem of this gyre is temporally and spatially variable^{5–7}. The causes of this variability are not well understood. Here we present evidence of a major change in the structure and productivity of the pelagic ecosystem in the subtropical North Pacific Ocean, an effect that we attribute to the 1991–92 El Niño–Southern Oscillation (ENSO) event. Decreased upper-ocean mixing and a change in circulation resulted in an increased abundance and activity of nitrogen-fixing microorganisms and a shift from a primarily nitrogen-limited to a primarily phosphorus-limited habitat with attendant changes in total and export production and in nutrient cycling pathways and rates.

As part of the US Joint Global Ocean Flux Study (JGOFS) and World Ocean Circulation Experiment (WOCE) programmes, we established a deep-water hydrostation⁸ (station ALOHA) ~100 km north of Oahu, Hawaii at 22° 45' N, 158° W to assess and interpret annual- to decadal-scale ecosystem variability in the North Pacific subtropical gyre. Research cruises, conducted at approximately monthly intervals, are designed to obtain oceanographic data on the physics, chemistry and biology of this representative oligotrophic North Pacific Ocean habitat. On each cruise, discrete water samples are collected from predetermined depths and analysed for inorganic and organic nutrients, photosynthetic pigments, suspended particulate matter, and particulate adenosine triphosphate (P-ATP). Rates of photosynthetic carbon fixation are measured using trace-metal-clean water sampling and *in situ* incubation techniques. Particle export from the euphotic zone is measured using drifting sediment traps. Water-column temperature, salinity and optical parameters are measured continuously with depth on each hydrocast. This comprehensive field experiment began in October 1988 and is expected to last for at least 10 years.

During the first four years of investigation, we observed a dramatic and sustained increase in total euphotic zone primary production that began in the spring of 1991 (Fig. 1a, b). We hypothesize that this productivity shift is a manifestation of the 1991–92 El Niño–Southern Oscillation (ENSO) event⁹. For comparative purposes, we divide our data set into pre-ENSO (1989–90) and ENSO (1991–92) intervals. Approximately 85% of the increased 114 g C m^{-2} in primary production during the 1991–92 period occurred in the upper 50 m of the water column (Fig. 1a, b and Table 1). This represents an increase in near-surface primary productivity of ~65% during the ENSO period. By comparison, primary production in the 100–200 m depth stratum did not change (Fig. 1c and Table 1). Furthermore, the higher productivity during 1991–92 is not the result of an increased standing stock of chlorophyll *a* or particulate carbon

TABLE 1 Oceanic habitat and ecosystem changes measured at station ALOHA (22° 45' N, 158° W)

Parameter*	1989–90 period†	1991–92 period	Significance‡
Total incident solar irradiance§ (MJ m ⁻²)	1.08 × 10 ⁴	1.02 × 10 ⁴	NS
Mean sea surface temperature (°C)	25.14 ± 1.36 (667)	25.16 ± 1.56 (655)	NS
Sea surface stability index* (°C)			
Ocean buoy no. 51001			
Mean	0.32 ± 0.23 (667)	0.38 ± 0.31 (655)	P < 0.001
Median	0.27	0.29	
Ocean buoy no. 51004			
Mean	0.26 ± 0.18 (694)	0.34 ± 0.32 (572)	P < 0.001
Median	0.22	0.24	
Surface-ocean dissolved nutrients (mg m ⁻²)			
Total dissolved nitrogen*	3,529 ± 456 (15)	4,000 ± 315 (21)	P < 0.001
Total dissolved phosphorus*	437 ± 43 (15)	481 ± 58 (21)	P < 0.01
Surface-ocean suspended particulate matter (mg m ⁻²)			
Chlorophyll a	4.5 ± 1.3 (17)	3.9 ± 1.1 (21)	NS
Carbon	1,290 ± 278 (14)	1,397 ± 266 (21)	NS
ATP	1.63 ± 0.31 (16)	1.30 ± 0.26 (19)	P < 0.01
Surface-ocean suspended particulate matter ratios**			
PC:PN (molar)			
Mean	8.3 ± 3.8 (13)	7.3 ± 1.4 (20)	NS
Median	7.1 (13)	6.9 (20)	
'Redfield'	6.6	6.6	
PC:PP (molar)			
Mean	112 ± 32 (14)	136 ± 24 (21)	P < 0.02
Median	111 (14)	135 (21)	
'Redfield'	106	106	
PN:PP (molar)			
Mean	14.2 ± 3.3 (13)	19.3 ± 5.2 (20)	P < 0.01
Median	13.6 (13)	18.7 (20)	
'Redfield'	16	16	
PC:P-ATP ratio (by weight)	822 ± 251 (14)	1,116 ± 300 (19)	P < 0.01
Primary production†† (mg C m ⁻² d ⁻¹)			
0–200 m	380 ± 92 (17)	534 ± 108 (18)	P < 0.001
0–50 m	202 ± 60 (17)	332 ± 78 (18)	P < 0.001
100–200 m	52 ± 27 (17)	41 ± 19 (18)	NS
Assimilation number (mg C per mg of chl a per h)	3.4 ± 1.5 (17)	5.9 ± 1.7 (18)	P < 0.001
Downward flux of particulate matter‡‡ (mg m ⁻² d ⁻¹)			
Carbon	35.5 ± 11.5 (18)	24.4 ± 9.5 (18)	P < 0.01
Nitrogen	4.99 ± 1.62 (18)	3.75 ± 1.53 (18)	P < 0.02
Phosphorus	0.49 ± 0.19 (18)	0.36 ± 0.15 (18)	P < 0.02
<i>Trichodesmium</i> abundance estimates§§			
% of total particulate carbon	5.8 ± 2.7 (9)	11.3 ± 3.3 (13)	P < 0.001
% of total chl a	8.1 ± 6.4 (9)	17.1 ± 7.0 (13)	P < 0.005

These changes were measured in the North Pacific subtropical gyre during 1989–92. The full data set is divided into two subsets for statistical comparison. All values are for the upper water column (0–50 m), except where otherwise noted. Depth-integrated values were calculated using the trapezoid rule.

* Values presented are mean ± 1 s.d. of the mean with number of observations during 1989–90 and 1991–92 (*n*) given in parentheses.

† Excluding HOT-9 data (see ref. 16).

‡ Significance values are based on a Wilcoxon two-sample test (ranked observations) of the null hypothesis stating that the two mean values were sampled from the same population of estimates. In cases where the null hypothesis could not be rejected at the $\alpha = 0.05$ significance level, the results are reported as not significant (NS). This is a conservative test and probability levels are higher than those estimated by parametric tests (for example, one-way analysis of variance, ANOVA). But differences found not to be significant by Wilcoxon's test were also NS by ANOVA. For the large data sets (sea surface temperature, stability index), a one-way ANOVA test was used.

§ Calculated from the monthly clear-sky surface irradiance values for 25° N latitude²⁵ corrected for hourly measurements of cloud coverage at the Honolulu International airport²⁶.

|| Measured at 1 m at the NDBC ocean buoy no. 51001 located at 23.4° N, 162.3° W.

¶ Defined as the daily excursions in sea surface temperature (SST), (maximum SST – minimum SST) with high positive values corresponding to calm ocean conditions; data are from NDBC ocean buoys nos 51001 (23.4° N, 162.3° W) and 51004 (17.4° N, 152.5° W).

* Sum of nitrate, nitrite, ammonium and organic nitrogen.

* Sum of phosphate, inorganic polyphosphate and organic phosphorus.

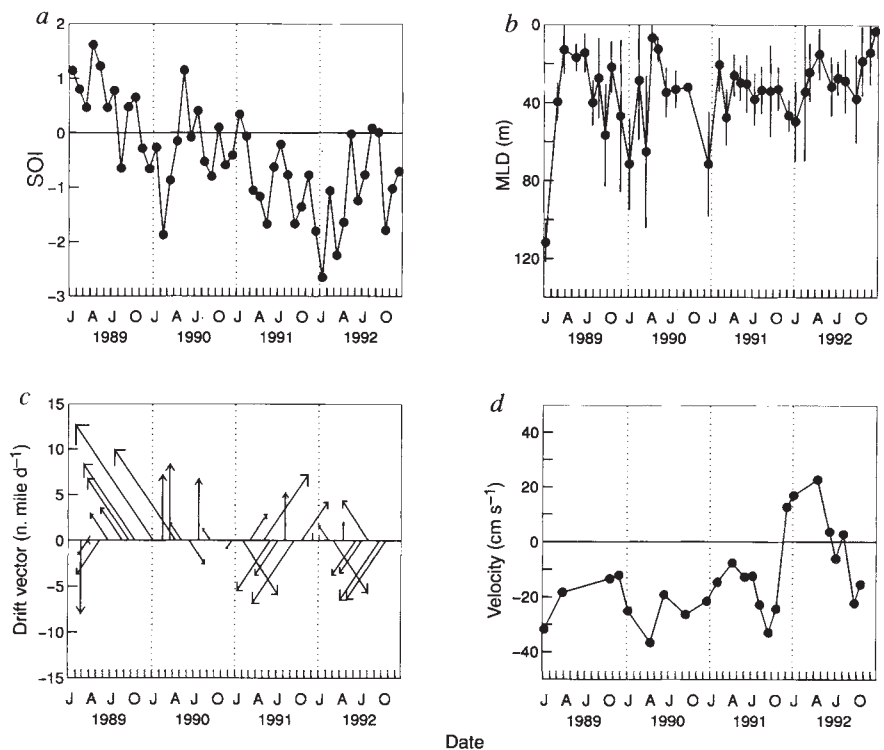
** PC, PN, PP and P-ATP are particulate carbon, nitrogen, phosphorus and adenosine triphosphate, respectively.

†† Measured by radiocarbon uptake using trace-metal-clean sampling and incubation methods.

‡‡ Measured using sediment traps deployed at the 150 m reference depth.

§§ Individual trichomes, containing an average of 100 cells, 50 ng C and 0.26 ng chl a, were concentrated by filtering 4–10 l of water onto a membrane filter and were enumerated on the basis of phycoerythrin autofluorescence under ultraviolet-illumination²⁷.

FIG. 2 a, The Troup Southern Oscillation index (SOI) defined as: $dP(\text{Tahiti}) - dP(\text{Darwin})/SD$, where $dP(\text{Tahiti})$ and $dP(\text{Darwin})$ are the monthly pressure anomalies (monthly mean minus the 1882–1991 mean) at Tahiti and Darwin, respectively, and SD is the monthly standard deviation of the difference²⁸. The SOI is highly correlated with the strength of the easterly trade winds that blow along the Equator. When SOI is positive, the trade winds are stronger than normal. Negative SOI values are associated with reduced trade winds, intensification of the North Equatorial countercurrent and increased flow of warm water into the eastern equatorial Pacific²⁹. Low SOI values generally indicate El Niño conditions. b, Upper ocean mixed-layer depth (MLD) at station ALOHA. The data presented are the mean ± 1 s.d. for each separate cruise, which typically averaged 15 discrete profiles. The MLD is defined as the depth where the density gradient first exceeds 5 g m^{-4} . c, Drift vectors, in nautical miles per day, for the sediment trap arrays deployed near station ALOHA on each Hawaii ocean time-series (HOT) cruise. The direction of the drift track is indicated by the orientation of the arrow and the velocity is given by the length of each arrow. The shift from a generally northwest drift in 1989, to a variable direction and lower speed drift in 1990–92 is one indication of a change in ocean circulation. d, Temporal variability in the velocity component of the Hawaiian ridge current in a direction 15° south of east as determined by hull-mounted acoustic Doppler current profiling techniques¹³. Averaging is from 21.7° to 22.6° N latitude and from 20–100 m. The negative of this value is an index of the strength of the Hawaiian ridge current³⁰.



1). Although apparently influential in nutrient cycling, *Trichodesmium* never becomes a numerically dominant species. This, too, is consistent with previous evidence that implicates *Trichodesmium* in the successional growth of non-nitrogen-fixing phytoplankton by a mechanism that may involve overproduction, and subsequent excretion, of fixed N as a normal metabolic process^{17,18}.

In the presence of large amounts of fixed N, the biological community dynamics are expected to shift towards phosphorus-limitation¹⁶. Although both dissolved nitrogen and dissolved phosphorus increased during the 1991–92 period, the molar N:P ratio of the increased nutrient pool was approximately twice that predicted by Redfield stoichiometry (16N:1P, ref. 10; Table 1). It should also be emphasized that the elevated C:P and N:P ratios of the particulate matter observed during 1991–92 (Table 1) are consistent with the predicted P “sparing effect” under phosphorus-limitation. Furthermore, the 30% decrease in particulate matter export in 1991–92 (see Fig. 1e and Table 1) is indicative of a more rapid recycling, extended residence time or perhaps accumulation of dissolved organic matter within the euphotic zone. The latter is predicted to occur during the growth of *Trichodesmium*¹⁹. If, as we present here, N_2 fixation contributes significantly to the oligotrophic Pacific Ocean nutrient budget then our conventional views of new (nitrate-based) and regenerated (ammonium-based) primary production may be in need of revision. It should also be noted that the potential sink for atmospheric carbon attributed to new production is quite different if the productivity is based on N_2 fixation rather than nitrate uptake.

The magnitude of the ecosystem changes observed at our subtropical study site is as large as the ENSO-induced effects observed previously for oceanic habitats thought to be directly in the dispersion wake of the warm water pool (Table 2). These changes include an enhancement of $\sim 50\%$ in primary production

for regions of the western Pacific²⁰, a 30–95% reduction in primary production in the eastern equatorial Pacific²¹, and a 25% and 69% reduction in nitrate and ammonium assimilation, respectively, for a station off Peru²².

ENSO events cause global changes in climate and can have significant and long-lasting effects on oceanic processes²³. Variability in ENSO strength and duration may lead to fundamentally different habitat and ecosystem changes. For the population and ecosystem changes that we report here, it is possible that event duration may be as important as event intensity.

The interpretation of low-frequency temporal variability in oligotrophic oceanic habits is limited by a lack of observations, especially for biogeochemical processes that must be assessed by discrete measurements. The data reported here, and those being collected at our sister JGOFS time-series station near Bermuda²⁴, are beginning to provide a revised view of the subtropical oceanic gyres. But unexpected ecosystem change potentially linked to ENSO events, as well as mesoscale ocean eddies and seasonal atmospheric dust inputs, suggests that comprehensive understanding of (and therefore ability to predict) oligotrophic ocean processes has not yet been achieved. □

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Evidence for a higher pH in the glacial ocean from boron isotopes in foraminifera

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RECORDS of past changes in the pH of the oceans should provide insights into how the carbonate chemistry of the oceans has changed over time. The latter is related to changes in the atmospheric CO₂ content, such as that which occurred during the last glacial–interglacial transition¹. Previous studies^{2,3} have shown that the fractionation of boron isotopes between sea water and precipitated carbonate minerals is pH-dependent. This finding has been used to reconstruct the evolution of ocean pH over the past 20 million years by analyses of boron isotopes in the carbonate shells of foraminifera⁴. Here we use the same approach to estimate changes in ocean pH between the last glacial and the Holocene period. We estimate that the deep Atlantic and Pacific oceans had a pH 0.3 ± 0.1 units higher during the last glaciation. The accompanying change in carbonate ion concentration is sufficient to account for the decrease in atmospheric pCO₂ during the glacial period¹. These results are consistent with the hypothesis⁵ that the low CO₂ content of the glacial atmosphere was caused by an increased ratio of organic carbon to carbonate in the ‘rain’ to the sea floor, which led to an increase in carbonate ion concentration (and thus in pH) of deep water without a corresponding increase in the lysocline depth.

The basis for the boron-isotope palaeoacidity indicator is that the uncharged species B(OH)₃ is enriched⁶ in ¹¹B by ~20‰ over the charged borate species, B(OH)₄⁻. As the fraction of the boron present as these species changes with pH, so also must their respective isotopic compositions. It is assumed that only the charged borate species is incorporated into carbonate minerals and that this incorporation occurs with a small and nearly constant isotopic fractionation. Thus the isotopic composition of

the borate incorporated into marine carbonate should serve as an indicator of the palaeoacidity. As the residence time of boron in sea water is many millions of years⁴, the boron isotope composition of sea water cannot have changed significantly over the past several glacial cycles. Thus the only mechanism by which the isotopic composition of the charged borate species in sea water can have varied is by changes in seawater pH.

We tested the validity of the boron isotope ratio as a pH indicator by plotting the δ¹¹B (defined in Table 1) of the modern planktonic and benthic foraminifera from both the Atlantic and Pacific oceans against the pH calculated from GEOSECS data⁷ at the sites of the cores from which they were obtained (Fig. 1). If we assume a small but constant fractionation (0.8‰) during the course of incorporation of B(OH)₄⁻ into the foraminiferal calcite, then it can be seen that the data is within analytical errors of the calculated curves. Within a particular pH range, the slope of the calculated curves is almost the same for both surface and deep water. Hence the difference in pH between two water masses of interest (ΔpH) calculated from Δδ¹¹B within this range of pH is similar for both the curves. (Δδ¹¹B = δ¹¹B_{SAMPLE a} - δ¹¹B_{SAMPLE b}.)

To exploit this potential, we analysed foraminiferal shells hand-picked from the coarse fraction of marine sediment. The shells were cleaned in hydrogen peroxide and then dissolved in hydrochloric acid; the solution was loaded directly onto a rhenium filament and analysed by negative-ion mass spectrometry³ at SUNY, Stony Brook. All sample solutions were analysed a number of times because this technique is subject to variable fractionation and isobaric interference⁸. Criteria for accepting an analysis include no indication of an isobaric interference on mass 42 (¹⁰BO₂⁻) from CNO (determined by monitoring mass 26, CN), and no variable fractionation during the analysis (mini-

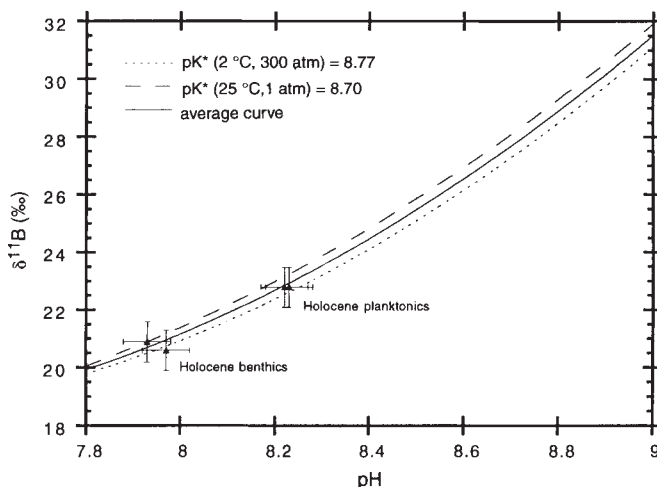


FIG. 1 Relationship between the boron isotope ratio of foraminifera and pH of sea water. The relationship was calculated by using an isotopic fractionation of 20‰ between the two aqueous species of boron⁶ and pressure-corrected pK* (where pK* is the apparent dissociation constant of boric acid in sea water) of the surface and deep sea water (Lyman¹⁶; Culberson and Pytkowicz¹⁷). It is assumed that B(OH)₄⁻ is the sole species being incorporated and that this incorporation takes place with a constant fractionation of 0.8‰. Based on this, the boron isotope composition of Holocene foraminifera would be expected to fall, within errors, on the two curves corresponding to the pH of sea water in which they grew. The Holocene planktonics and the benthics have been plotted against the GEOSECS-calculated⁷ pH of surface and deep water (3,000 m), respectively, at the sites of the cores from which the samples were obtained. To simplify calculations we adopt an average curve, drawn between the two curves. Based on this average curve, ΔpH has been calculated from Δδ¹¹B (ΔpH = 0.29(Δδ¹¹B) - 0.0037(Δδ¹¹B)(δ¹¹B_a + δ¹¹B_b), where subscripts a and b represent two different samples).